File Guide

SubSplit — escrow-protected subscription sharing

DOTE 6751 Database Systems · Group Project · November 2026

This document is the map of the package: every file, what it is for, how to rebuild the database, how to verify it, and which file to hand over when someone asks.

1. The map

Generated by scanning the folder, so the list and the files agree.

```
数据库大作业_SubSplit/
  README.md   (2 KB)
  File_Guide_文件阅读指南/
    File_Guide.docx   (42 KB)
    File_Guide.pdf   (210 KB)
    File_Guide_CN.docx   (43 KB)
    File_Guide_CN.pdf   (452 KB)
  PresentationSlides/
    Presentation_Script.md   (4 KB)
    QA_Prep.md   (4 KB)
    SubSplit.pdf   (579 KB)
    SubSplit.pptx   (510 KB)
    SubSplit_Narrative.docx   (41 KB)
    SubSplit_Narrative.pdf   (128 KB)
    Task_Allocation.md   (3 KB)
    SubSplit_ChineseVersion/
      SubSplit_CN.pdf   (1097 KB)
      SubSplit_CN.pptx   (513 KB)
  Project_Spec_产品介绍/
    Product_Spec_CN.docx   (35 KB)
    Product_Spec_CN.md   (9 KB)
    Product_Spec_EN.docx   (41 KB)
    Product_Spec_EN.md   (9 KB)
  design/
    Constraint_Inventory.md   (5 KB)
    Design_Rationale.md   (8 KB)
    Entity_Relationship_Catalog.md   (7 KB)
    Normalization_Proof.xlsx   (12 KB)
    Relational_Schema.md   (6 KB)
    Relationship_Matrix.xlsx   (7 KB)
    Requirements.md   (5 KB)
  erd/
    ERD_conceptual.png   (139 KB)
    ERD_conceptual.svg   (77 KB)
    ERD_full.drawio   (7 KB)
    ERD_logical_full.png   (245 KB)
    ERD_logical_full.svg   (155 KB)
    ERD_module_escrow.png   (138 KB)
    ERD_module_escrow.svg   (87 KB)
    ERD_module_group.png   (98 KB)
    ERD_module_group.svg   (72 KB)
    ERD_module_user.png   (103 KB)
    ERD_module_user.svg   (71 KB)
    flow_escrow.png   (96 KB)
```

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  flow_escrow.svg   (55 KB)
query_results/
  _summary.json   (5 KB)
  q01_result.csv   (<1 KB)
  q01_result.png   (30 KB)
  q02_result.csv   (<1 KB)
  q02_result.png   (28 KB)
  q03_result.csv   (<1 KB)
  q03_result.png   (56 KB)
  q04_result.csv   (<1 KB)
  q04_result.png   (52 KB)
  q05_result.csv   (<1 KB)
  q05_result.png   (62 KB)
  q06_result.csv   (<1 KB)
  q07_result.csv   (<1 KB)
  q08_result.csv   (<1 KB)
  q09_result.csv   (<1 KB)
  q10_result.csv   (<1 KB)
  q11_result.csv   (<1 KB)
  q11_result.png   (45 KB)
  q12_result.csv   (<1 KB)
  q13_result.csv   (<1 KB)
  q14_result.csv   (<1 KB)
  q15_result.csv   (<1 KB)
  q16_result.csv   (<1 KB)
  q17_result.csv   (<1 KB)
  q17_result.png   (53 KB)
  q18_result.csv   (<1 KB)
sql/
  00_run_all.sql   (<1 KB)
  01_schema.sql   (13 KB)
  02_seed.sql   (266 KB)
  03_queries.sql   (12 KB)
  04_views_triggers.sql   (6 KB)
  05_transactions.sql   (4 KB)
  06_checks.sql   (5 KB)
  subsplit_db_full.sql   (270 KB)
```

2. Every file, and what it is for

Grouped by folder. What matters is when you would open it.

Root

File	Size	What it is	What is inside
README.md	2 KB	Quick orientation	Three sentences on what the package is and where to start.

File_Guide_文件阅读指南/

File	Size	What it is	What is inside
File_Guide.docx	42 KB	—	—
File_Guide.pdf	210 KB	—	—
File_Guide_CN.docx	43 KB	—	—
File_Guide_CN.pdf	452 KB	—	—

PresentationSlides/

File	Size	What it is	What is inside
Presentation_Script.md	4 KB	Speaking script	What to say on each slide, with timings and speaker split.
QA_Prep.md	4 KB	Q&A preparation	Likely questions with answers short enough to say aloud.
SubSplit.pdf	579 KB	Deck PDF	For sending or opening without PowerPoint.
SubSplit.pptx	510 KB	Deck (EN)	Fourteen slides, speaker notes on every one.
SubSplit_Narrative.docx	41 KB	Narrative	About 1,900 words of continuous prose across the same six sections.
SubSplit_Narrative.pdf	128 KB	Narrative PDF	The same text, fixed layout.
Task_Allocation.md	3 KB	Task allocation	Seven technical tasks with difficulty and hours; no presentation role.

PresentationSlides\SubSplit_ChineseVersion/

File	Size	What it is	What is inside
SubSplit_CN.pdf	1097 KB	Deck PDF (CN)	PDF of the Chinese deck.
SubSplit_CN.pptx	513 KB	Deck (CN)	The same fourteen slides in Chinese.

Project_Spec_产品介绍/

File	Size	What it is	What is inside
Product_Spec_CN.docx	35 KB	Product spec (CN)	Positioning, vision, substitutes, scenarios, features, economics, metrics, risks, roadmap.
Product_Spec_CN.md	9 KB	Product spec source (CN)	The same document as Markdown.
Product_Spec_EN.docx	41 KB	Product spec (EN)	The same document in English.
Product_Spec_EN.md	9 KB	Product spec source (EN)	The same document as Markdown.

design/

File	Size	What it is	What is inside
Constraint_Inventory.md	5 KB	Constraint inventory	Every rule, its type, where it is enforced, and what happens when it breaks.
Design_Rationale.md	8 KB	Design rationale	Why these tables, why the weak entity, why derived values are stored, what was rejected.
Entity_Relationship_Catalog.md	7 KB	E-R catalogue	Every entity and relationship written out: cardinality, participation, attributes.
Normalization_Proof.xlsx	12 KB	Normalization proof	Functional dependencies, candidate keys and the 3NF/BCNF

			argument, per table.
Relational_Schema.md	6 KB	Relational schema	The E-R design translated into relations, with keys and mapping rules.
Relationship_Matrix.xlsx	7 KB	Relationship matrix	Cardinality, participation and referential action for every relationship.
Requirements.md	5 KB	Requirements and scope	The problem, the actors, the functional requirements, scope and assumptions.

erd/

File	Size	What it is	What is inside
ERD_conceptual.png	139 KB	Conceptual E-R	Entities, relationship diamonds and cardinality.
ERD_conceptual.svg	77 KB	Vector version	Scales without blur; opens in a browser.
ERD_full.drawio	7 KB	Editable source	A draw.io file for hand editing.
ERD_logical_full.png	245 KB	Logical schema	Every table with columns, keys and connectors.
ERD_logical_full.svg	155 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_escrow.png	138 KB	Module E-R	A zoomed view of that module.
ERD_module_escrow.svg	87 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_group.png	98 KB	Module E-R	A zoomed view of that module.
ERD_module_group.svg	72 KB	Vector version	Scales without blur; opens in a browser.
ERD_module_user.png	103 KB	Module E-R	A zoomed view of that module.
ERD_module_user.svg	71 KB	Vector version	Scales without blur; opens in a browser.
flow_escrow.png	96 KB	Money-flow figure	From member to escrow, then monthly to the host.
flow_escrow.svg	55 KB	Vector version	Scales without blur; opens in a browser.

query_results/

File	Size	What it is	What is inside
_summary.json	5 KB	Query summary	Rows, timing and columns per query.
q01_result.csv	<1 KB	Query 1 result	The full output of that query on the seeded instance.
q01_result.png	30 KB	Query 1 result image	The result rendered as a table image.
q02_result.csv	<1 KB	Query 2 result	The full output of that query on the seeded instance.
q02_result.png	28 KB	Query 2 result image	The result rendered as a table image.
q03_result.csv	<1 KB	Query 3 result	The full output of that query on the seeded instance.
q03_result.png	56 KB	Query 3 result image	The result rendered as a table image.
q04_result.csv	<1 KB	Query 4 result	The full output of that query on the seeded instance.
q04_result.png	52 KB	Query 4 result image	The result rendered as a table image.
q05_result.csv	<1 KB	Query 5 result	The full output of that query on the seeded instance.

q05_result.png	62 KB	Query 5 result image	The result rendered as a table image.
q06_result.csv	<1 KB	Query 6 result	The full output of that query on the seeded instance.
q07_result.csv	<1 KB	Query 7 result	The full output of that query on the seeded instance.
q08_result.csv	<1 KB	Query 8 result	The full output of that query on the seeded instance.
q09_result.csv	<1 KB	Query 9 result	The full output of that query on the seeded instance.
q10_result.csv	<1 KB	Query 10 result	The full output of that query on the seeded instance.
q11_result.csv	<1 KB	Query 11 result	The full output of that query on the seeded instance.
q11_result.png	45 KB	Query 11 result image	The result rendered as a table image.
q12_result.csv	<1 KB	Query 12 result	The full output of that query on the seeded instance.
q13_result.csv	<1 KB	Query 13 result	The full output of that query on the seeded instance.
q14_result.csv	<1 KB	Query 14 result	The full output of that query on the seeded instance.
q15_result.csv	<1 KB	Query 15 result	The full output of that query on the seeded instance.
q16_result.csv	<1 KB	Query 16 result	The full output of that query on the seeded instance.
q17_result.csv	<1 KB	Query 17 result	The full output of that query on the seeded instance.
q17_result.png	53 KB	Query 17 result image	The result rendered as a table image.
q18_result.csv	<1 KB	Query 18 result	The full output of that query on the seeded instance.

sql/

File	Size	What it is	What is inside
00_run_all.sql	<1 KB	Rebuild script	Calls the other scripts in the only correct order.
01_schema.sql	13 KB	DDL	Creates the database and every table with keys, checks and indexes.
02_seed.sql	266 KB	Seed data	Deterministic data; ends with a reconciliation statement.
03_queries.sql	12 KB	Eighteen queries	Three families; each statement runs on its own.
04_views_triggers.sql	6 KB	Views and triggers	Read models and the triggers that enforce business rules.
05_transactions.sql	4 KB	Transaction episodes	Self-contained episodes, rolled back, so the data is unchanged.
06_checks.sql	5 KB	Acceptance checks	Twelve checks; every row must read PASS.
subsplit_db_full.sql	270 KB	Full dump	Schema, data, views and triggers in one file.

3. How to rebuild the database

In Navicat: right-click the connection → Run SQL File → sql/subsplit_db_full.sql.

Or step by step: 01_schema → 02_seed → 04_views_triggers → 06_checks → 03_queries.

Command line: mysql -u root -p < 00_run_all.sql

The order matters: the seed contains historical states that the live capacity trigger would rightly refuse.

4. How to verify

Eight steps. The first six take three minutes; the last two prove the constraints actually bite.

#	Check	How	A pass looks like
1	Service running	MySQL80 is Running	Started
2	Client connects	Test Connection	Success
3	Schema complete	Restore and expand	11 tables, 2 views, 5 triggers
4	Data arrived	COUNT(*) on five tables	300 / 60 / 300 / 300 / 2,172
5	Acceptance script	Run 06_checks.sql	12 rows, all PASS
6	Queries answer	Run 03_queries.sql	q01 two rows, q05 six, q11 ten
7	Constraints refuse	Run the three statements below	Each fails, nothing inserted
8	Money reconciles	SUM(released) vs ledger SUM(amount)	The two are equal

These three are meant to fail. If one succeeds, a constraint is missing.

Overfill a group

```
INSERT INTO membership_seat (group_id, member_id, joined_at, expires_at)
VALUES (2, 2, NOW(), NOW() + INTERVAL 1 YEAR);
```

Expected: error 1644, capacity exceeded

Pay a month twice

```
INSERT INTO monthly_payout_log (escrow_id, release_month, amount, platform_fee)
VALUES (1, '2026-09-01', 10.00, 0.50);
```

Expected: error 1062, duplicate key

Break the balance

```
UPDATE escrow_payment SET remaining_amount = 0.01 WHERE escrow_id = 1;
```

Expected: error 3819, CHECK violated

5. What was checked before delivery

Twelve checks, all executed. Any FAIL is a defect.

#	Check	Violations	Result
C01	Eleven tables present	11	PASS
C02	Two views and five triggers	7	PASS
C03	No group over capacity	0	PASS
C04	Balance identity holds	0	PASS
C05	Balance equals ledger	0	PASS
C06	No escrow over 12 releases	0	PASS
C07	No month paid twice	0	PASS
C08	One default instrument each	0	PASS
C09	Credit score matches ledger	0	PASS

C10	Seats end after they start	0	PASS
C11	Reviewed disputes have a reviewer	0	PASS
C12	Vault timestamp matches journal	0	PASS

6. Decisions worth knowing

Eleven tables, not seven

The extra four exist because a value is not enough: complaints, credential history, instruments and credit events each need rows.

The seat ceiling lives in the product

Copying it into every group would create a transitive dependency; the trigger joins to read it.

Four derived values are stored

Released, remaining, credit score and full_at — each guarded by a CHECK or a trigger, audited by Q11.

Refunds are manual

Automatic refunds are gameable; the database records, freezes and queues, a human decides.

The e-mail column was removed

A synthetic instance cannot carry a believable institutional address; keeping it was a tell.